

Enhanced remote sensing and deep learning aided water quality detection in the Ganges River, India supporting monitoring of aquatic environments

Lavanya Kandasamy^a, Anand Mahendran^{b,*}, Sai Harsha Varma Sangaraju^a, Preksha Mathur^a, Soham Vijaykumar Faldu^a, Manuel Mazzara^c

^a School of Computer Science and Engineering, VIT, Vellore 632014, Tamil Nadu, India

^b School of Computer Science and Engineering VIT, Chennai 600127, Tamil Nadu, India

^c Institute of Software Development and Engineering, Innopolis University, Innopolis, Russia

ARTICLE INFO

Keywords:

Chlorophyll-a monitoring
Deep learning
Environmental remote sensing
Google earth engine
Machine learning models
Remote water quality assessment
Satellite imagery analysis
Time series forecasting
Water quality parameters

ABSTRACT

Water pollution is a pressing global concern, impacting numerous communities across the world. Existing water quality monitoring systems rely on static or periodically collected data, presenting limitations in their ability to provide real-time dynamic insights. This research introduces an innovative approach to address this gap—a dynamic data intake system capable of identifying contamination sources, employing remote sensing techniques to track temporal changes, and issuing timely alerts for safeguarding crucial water resources. The proposed system adopts a hybrid methodology, integrating the QAA-v5 algorithm to derive essential parameters. These parameters serve as input for a pre-trained CatBoost model, which facilitates real-time calculations of chlorophyll-a concentrations at specified geographical coordinates. For future forecasting, the system leverages two distinct models: NBeats and CatBoost Time-Series. Notably, the CatBoost model achieves a commendable regression score of 0.985. For a comprehensive assessment and validation of the system's performance, the research draws upon the dataset provided by the International Ocean-Color Coordinating Group (IOCCG). The innovative framework introduced in this study exhibits considerable promise in advancing water quality protection and monitoring, making a significant contribution to the field of environmental research and management.

1. Introduction

WATER a life-sustaining resource, faces relentless pollution from human activities, especially in remote areas lacking water quality monitoring [1]. Remote sensing technology combined with deep learning algorithms, like CNN, ResNet, and many other models, offers a potent solution. Water quality parameters, including chlorophyll-a, are crucial for assessing water quality (Guimarães [2]). The significance of chlorophyll-a monitoring extends beyond basic water quality assessment, serving as a key indicator of eutrophication and potential harmful algal blooms that can severely impact aquatic ecosystems and human

health.

These parameters encompass a wide spectrum of measurements, including chemical oxygen demand (COD), biological oxygen demand (BOD), total suspended solids (TSS), total dissolved solids (TDS), pH, temperature, turbidity, dissolved oxygen (DO), and nutrients such as phosphates and nitrates. Chlorophyll-a maintains particular significance in water quality assessment, as demonstrated by Summers [3]. The ability to monitor these parameters continuously and accurately is crucial for early warning systems and environmental protection strategies.

Remote sensing of chlorophyll-a provides invaluable capabilities for

Abbreviations: BOD, Biological Oxygen Demand; COD, Chemical Oxygen Demand; CNN, Convolutional Neural Network; DO, Dissolved Oxygen; DOM, Dissolved Organic Matter; GIS, Geographic Information Systems; IOCCG, International Ocean-Color Coordinating Group; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error; ML, Machine Learning; MSE, Mean Squared Error; NOAA, National Oceanic and Atmospheric Administration; OLI, Operational Land Imager; QAA-v5, Quasi-Analytical Algorithm Version 5; RNN, Recurrent Neural Network; TIRS, Thermal Infrared Sensor; TSS, Total Suspended Solids; TDS, Total Dissolved Solids; TSI, Trophic State Index; WQI, Water Quality Index; XGBoost, eXtreme Gradient Boosting.

* Corresponding author.

E-mail addresses: lavanya.k@vit.ac.in (L. Kandasamy), manand@vit.ac.in (A. Mahendran), saiharsha.varma2020@vitstudent.ac.in (S.H.V. Sangaraju), preksha.mathur2020@vitstudent.ac.in (P. Mathur), m.mazzara@innopolis.ru (M. Mazzara).

<https://doi.org/10.1016/j.rineng.2024.103604>

Received 6 September 2024; Received in revised form 24 November 2024; Accepted 1 December 2024

Available online 13 December 2024

2590-1230/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

monitoring, detecting water quality issues, and tracking changes in a water body's trophic state. The Trophic State Index (TSI) serves as a critical metric for assessing nutrient and productivity levels, with chlorophyll-a serving as a primary indicator. Recent advancements in satellite technology have significantly enhanced the resolution and accuracy of remote sensing data, enabling more precise monitoring of water quality parameters. According to Elhag et al. [4], remote-sensing technologies have demonstrated remarkable cost-effectiveness and efficiency in water quality monitoring, achieving accuracy rates up to 92 %. This technology has already been integrated into operations by over 50 % of water management entities [5], highlighting its growing acceptance and reliability in the field of environmental monitoring.

Recent research by Yang et al. [6] highlights how the proliferation of satellites and advancements in camera technology have substantially enhanced the accessibility of remote-sensing data for water quality monitoring. The evolution of satellite sensors has led to improved spatial and temporal resolution, making it possible to detect subtle changes in water quality parameters. While traditional empirical methods have been conventionally employed for converting satellite reflectance values into meaningful water quality parameters, opportunities exist for global optimization of these approaches. The integration of machine learning techniques, particularly deep learning algorithms, represents a significant advancement in improving the accuracy and reliability of satellite-based water quality monitoring. The integration of deep learning with remote sensing data represents a significant advancement in environmental monitoring technology. This combination enables automated feature extraction, pattern recognition, and anomaly detection, capabilities that are particularly valuable in water quality monitoring. The system's ability to process and analyze large volumes of satellite data in real-time, while maintaining high accuracy levels, addresses a critical gap in current monitoring approaches. Wang et al. [7] found that such integrated systems can reduce monitoring costs by up to 60 % while providing coverage of previously inaccessible water bodies, making them particularly valuable for developing regions.

This paper presents a dynamic water quality detection system that integrates deep learning with remote sensing capabilities. The system comprises a mobile application interface for real-time interaction, an API infrastructure for data processing, and integration with Google Earth Engine for data retrieval. The implementation includes chlorophyll-a estimation algorithms with a three-month analysis capability and user-specific model adaptation. The architecture employs dual models, CatBoost and NBeats, along with automated alert mechanisms. The current implementation focuses on ten locations along the Ganga River, one of the world's most significant and stressed waterways, with built-in capacity for geographical expansion. The Ganga River system, supporting over 400 million people, serves as an ideal case study for demonstrating the effectiveness of our approach in monitoring complex water systems.

The remainder of this paper is organized as follows: Section II reviews existing literature in the field. Section III details the methodology for data extraction and preprocessing. Section IV presents the implementation of various machine learning models. Section V discusses the results and their implications. Finally, Section VI concludes with recommendations for future research directions.

2. Literature survey

The intricacies of water quality assessment through remote sensing are underscored by the complexities of converting satellite reflectance values into meaningful parameters like chlorophyll-a concentration. The process of extracting and converting parameters from satellite imagery presents a formidable challenge within the domain of water quality remote sensing. Satellite images primarily provide reflectance values at different wavelengths, which serve as the fundamental data source. To translate these reflectance values into essential parameters like chlorophyll-a concentration, a series of intricate mathematical

transformations and algorithms are necessitated. This conversion process is intricate and entails multiple steps, relying on algorithms derived from bio-optical models and empirical relationships between reflectance values and chlorophyll-a concentrations [7]. Notably, one of the inherent complexities lies in the reliance on assumptions that may not universally hold, especially for higher chlorophyll-a values, potentially resulting in significant estimation errors.

The accuracy of predicting chlorophyll-a values from reflectance values through the application of complex mathematical formulae is inherently limited. The limitation arises because mathematical formulae struggle to accurately represent the intricate and non-linear relationship between reflectance values and chlorophyll-a concentrations. The inherent complexity of water quality dynamics, combined with variations in different environmental conditions, makes it challenging for traditional mathematical models to capture and generalize these relationships effectively. A viable solution to enhance accuracy is found in the realm of machine learning and deep learning models. These models possess the capability to glean intricate relationships between reflectance values and chlorophyll-a concentrations from extensive datasets, offering a more precise and reliable means of prediction. For instance, machine learning techniques such as XGBoost, as introduced by Chen, and Guestrin, [8], or deep learning methods such as Recurrent Neural Networks (RNN) as proposed by Ye, et al. [9], can be employed. Successful implementation of these techniques hinges upon the size and quality of the training dataset. To ensure optimal performance and generalization of new data, a large and diverse dataset becomes imperative.

In addition, this paper delves into the utilization of the N-Beats model, as outlined by Oreshkin et al. [10], for the prediction of chlorophyll-a values based on reflectance data. This deep learning model, expressly designed for time series analysis, exhibits the capacity to handle multi-step ahead predictions, taking into consideration previous time steps and seasonality in the data. The ensuing results of this endeavour are discussed comprehensively.

This paper utilizes machine learning techniques, Nasir [11] compares the performance of CatBoost and XGBoost algorithms for water quality prediction. It provides insights into their strengths and weaknesses in predicting various water quality parameters [11]. While existing WQI models struggle with consistent classification due to varied schemes, Uddin [12] utilizes three ML models and XGBoost classifier to achieve near-perfect accuracy in predicting coastal water quality which solidifies our work.

Numerous research publications align closely with the objectives of this study. They highlight the importance of water quality monitoring and propose various approaches to address the limitations of existing systems. Dutta et al. [13] discusses the use of data mining techniques, such as artificial neural networks (ANN) and geographic information systems (GIS), for classifying water pollutants and analyzing their impact. Dhumvad et al. [14] presents a decision support system that integrates Internet of Things, machine learning, and embedded systems to predict water pollution using physical, chemical, and biological parameters. For instance, Sun, et al. [15] introduced proximal remote sensing as a methodology for evaluating water quality. However, a notable drawback of this approach lies in the requirement for the installation of detection equipment at individual water sources. In contrast, our approach harnesses the extensive capabilities of Google Earth Engine data in assessing water quality, circumventing the need for physical installations at each unique water source. This amalgamation facilitated the development of a remote sensing algorithm tailored for pollutant source identification based on the optical and fluorescence properties of DOM. However, it's imperative to note that this approach has a limitation in its reliance on in-situ sample data for PS dataset creation and summated in Table 1.

Table 1
Literature analysis.

Study	Methodology	Key Findings
Dutta et al. [13]	Data mining techniques for classification	Use of artificial neural networks (ANN) and geographic information systems (GIS) for water pollutant classification.
Dhumvad et al. [14]	IoT and machine learning for prediction	Integration of IoT and machine learning for water pollution prediction.
Sun, X., et al. [15]	Proximal remote sensing	Introduction of proximal remote sensing for water quality evaluation.
Oreshkin et al. [10]	N-Beats model for time series analysis	Application of the N-Beats model for predicting chlorophyll-a values based on reflectance data.
Chen, T., & Guestrin, C. [8]	XGBoost for predictive modelling	Introduction and application of the XGBoost algorithm for predictive modelling in various domains.
Ye, Q., et al. [9]	Recurrent Neural Networks (RNN)	Proposal and application of Recurrent Neural Networks (RNN) for predictive modelling.
Nasir, N. (2022) [11]	Machine learning techniques for water quality prediction	Comparison of CatBoost and XGBoost algorithms for water quality prediction, providing insights into their strengths and weaknesses in predicting various water quality parameters.
Uddin. [12]	Machine learning models and XGBoost classifier for coastal water quality prediction	Utilization of three ML models and an XGBoost classifier to achieve near-perfect accuracy in predicting coastal water quality, reinforcing the effectiveness of machine learning in this domain.

Table 2
IOCCG Parameters Description.

Sr. No.	Parameter	Description
1	Absorption coefficient of phytoplankton pigments ($a_{ph}(\lambda)$)	The rate at which light is absorbed by phytoplankton pigments. This is a critical parameter because it aids in estimating the amount of phytoplankton in water.
2	Absorption coefficient of coloured dissolved organic matter	The rate at which light is absorbed by organic matter in water. This is a critical parameter because it aids in estimating the amount of organic matter in water.
3	Absorption coefficient of detritus/mineral ($a_{dg}(\lambda)$)	The rate at which light is absorbed by detritus or mineral particles in water. This is an important parameter because it helps estimate the amount of these particles in water.
4	Absorption coefficient of the total ($a(\lambda)$)	The total rate of light absorption in water. This parameter is the sum of phytoplankton pigment absorption coefficients, coloured dissolved organic matter, and detritus/mineral absorption coefficients.
5	Backscattering coefficient of phytoplankton ($b_{bp}(\lambda)$)	The rate at which light is scattered back in the opposite direction of the light source by phytoplankton. This is a critical parameter because it aids in estimating the number of phytoplankton in water.
6	Backscattering coefficient of detritus/mineral ($b_{dg}(\lambda)$)	The rate at which light is scattered back in the opposite direction by detritus or mineral particles in water. This is an important parameter because it helps estimate the amount of these particles in water.
7	Backscattering coefficient of the total ($b_b(\lambda)$)	The total rate at which light is scattered back into the water. This parameter is the sum of the phytoplankton and detritus/mineral backscattering coefficients.
8	Below-surface remote-sensing reflectance, $Lu(0-)/Ed(0-)$ ($r_{rs}(\lambda)$)	The amount of light reflected from the water surface and detected by remote sensors below the water surface. This is an important parameter for water property remote sensing.
9	Above-surface remote-sensing reflectance, $Lw(0+)/Ed(0+)$ ($R_{rs}(\lambda)$)	The amount of light reflected from the water surface and detected by remote sensors above the water surface. This is an important parameter for water property remote sensing.
10	Below-surface irradiance reflectance, $Eu(0-)/Ed(0-)$	The proportion of upward radiance at a given depth to downward irradiance just below the water's surface. This parameter can be used to investigate light penetration in water.
11	Downwelling diffuse attenuation coefficient of downwelling irradiance at depth z , with z set at 0, 5, 10, 20, and 40 m from the surface	The rate at which light is attenuated as it descends through the water column to a given depth. This is a critical parameter because it aids in estimating the amount of light available to aquatic organisms at various depths.
12	Absorption of pure seawater ($a_w(\lambda)$)	The rate at which light is absorbed by pure seawater in the absence of particles or organic matter. This is an important parameter for correcting the effects of water on other optical property measurements.
13	Backscattering coefficient of pure seawater ($b_{bw}(\lambda)$)	The rate at which light is scattered back in the opposite direction of the light source by pure seawater that contains no particles or organic matter. This is an important parameter for correcting the effects of water on other optical property measurements.

3. Dataset

In this section, the research delves into the dataset employed for the purpose of training, validation, and evaluation. The dataset's intrinsic composition, origins, and relevance to the study's objectives are meticulously presented below.

3.1. International ocean color coordinating group

The International Ocean Colour Coordinating Group (IOCCG) (<http://ioccg.org>) (30 November 2022) dataset stands as a seminal resource in the domain of ocean colour remote sensing applications. It encapsulates a comprehensive array of measurements pertaining to critical ocean colour parameters, including but not limited to chlorophyll concentration, backscattering coefficient, and absorption coefficients. This dataset contains a variety of parameters used to train a hybrid model. The parameters pertaining to water's optical properties

and how they interact with light. These parameters are of paramount significance in the context of understanding the intricate biogeochemical properties of the ocean [16] and its pivotal role in the carbon cycle. The details of the parameters are as follows that are represented in Table 2:

- **Absorption coefficient of phytoplankton pigments ($a_{ph}(\lambda)$):** The rate at which light is absorbed by phytoplankton pigments. This is a critical parameter because it aids in estimating the number of phytoplankton in water.
- **Absorption coefficient of coloured dissolved organic matter:** The rate at which light is absorbed by organic matter in water. This is a critical parameter because it aids in estimating the amount of organic matter in water.
- **Absorption coefficient of detritus/mineral ($a_{dg}(\lambda)$):** The rate at which light is absorbed by detritus or mineral particles in water. This

is an important parameter because it helps estimate the amount of these particles in water.

- **Absorption coefficient of the total ($a(\lambda)$):** The total rate of light absorption in water. This parameter is the sum of phytoplankton pigment absorption coefficients, coloured dissolved organic matter, and detritus/mineral absorption coefficients.
- **Backscattering coefficient of phytoplankton ($b_{bp}(\lambda)$):** The rate at which light is scattered back in the opposite direction of the light source by phytoplankton. This is a critical parameter because it aids in estimating the number of phytoplankton in water
- **Backscattering coefficient of detritus/mineral ($b_{dg}(\lambda)$):** The rate at which light is scattered back in the opposite direction by detritus or mineral particles in water. This is an important parameter because it helps estimate the amount of these particles in water.
- **Backscattering coefficient of the total ($b_b(\lambda)$):** The total rate at which light is scattered back into the water. This parameter is the sum of the phytoplankton and detritus/mineral backscattering coefficients.
- **Below-surface remote-sensing reflectance, $Lu(0-)/Ed(0-)$ ($r_{rs}(\lambda)$):** The amount of light reflected from the water surface and detected by remote sensors below the water surface. This is an important parameter for water property remote sensing.
- **Above-surface remote-sensing reflectance, $Lw(0+)/Ed(0+)$ ($R_{rs}(\lambda)$):** The amount of light reflected from the water surface and detected by remote sensors above the water surface. This is an important parameter for water property remote sensing.
- **Below-surface irradiance reflectance, $Eu(0-)/Ed(0-)$:** The proportion of upward radiance at a given depth to downward irradiance just below the water's surface. This parameter can be used to investigate light penetration in water.
- **Downwelling diffuse attenuation coefficient of downwelling irradiance at depth z , with z set at 0, 5, 10, 20, and 40 m from the surface:** The rate at which light is attenuated as it descends through the water column to a given depth. This is a critical parameter because it aids in estimating the amount of light available to aquatic organisms at various depths.
- **Absorption of pure seawater ($a_w(\lambda)$):** The rate at which light is absorbed by pure seawater in the absence of particles or organic matter. This is an important parameter for correcting the effects of water on other optical property measurements.
- **Backscattering coefficient of pure seawater ($b_{bw}(\lambda)$):** The rate at which light is scattered back in the opposite direction of the light source by pure seawater that contains no particles or organic matter. This is an important parameter for correcting the effects of water on other optical property measurements.

The distribution of chlorophyll-*a* data can be visually portrayed through the utilization of box plots and scatter plots. Additionally, a heatmap proves invaluable in uncovering potential interrelationships among the various parameters present in the dataset.

Notably, within the scope of the present study, the IOCCG dataset served as the foundation for the training of a sophisticated model. The analysis of the IOCCG Data frame is given in Table 3. This model was intended to facilitate the real-time detection and quantification of water quality parameters within the revered Ganga River, one of India's most prominent water bodies. Over the course of time, the Ganga River has faced a lamentable degradation owing to anthropogenic factors such as industrialization, urbanization, and agricultural activities.

Subsequent to the training process, the model was subjected to rigorous real-time evaluation along the length of the Ganga River, facilitating the measurement of crucial water quality parameters, including chlorophyll concentration. This real-time monitoring capability emerges as a pivotal component for the effective management and conservation of water resources.

To bolster the reliability and veracity of the model's output, NOAA (National Oceanic and Atmospheric Administration) data was enlisted

Table 3

Analysis of the IOCCG data frame.

Sr	Column	Non-Null Count	Dtype
0	wavelength	20,500 non-null	float64
1	Aw	20,500 non-null	float64
2	Bw	20,500 non-null	float64
3	C	20,500 non-null	float64
4	Sg	20,500 non-null	float64
5	Sd	20,500 non-null	float64
6	a_ph	20,500 non-null	float64
7	a_g	20,500 non-null	float64
8	a_dm	20,500 non-null	float64
9	A	20,500 non-null	float64
10	bb_ch	20,500 non-null	float64
11	bb_dm	20,500 non-null	float64
12	Bb	20,500 non-null	float64
13	Rrs	20,500 non-null	float64
14	r_rs	20,500 non-null	float64

for validation purposes. NOAA, a preeminent scientific agency within the United States, specializes in monitoring and cataloguing the dynamic conditions of oceans, major water bodies, and the Earth's atmospheric conditions. This data resource is a pivotal tool in substantiating the accuracy and efficacy of models dedicated to ocean and water quality monitoring.

3.2. Landsat - 8

Landsat-8, launched in 2013, constitutes an Earth observation satellite under the aegis of the United States Geological Survey (USGS) (<https://landsat.gsfc.nasa.gov/satellites/landsat-8/>) (30 November 2022), Tavora et al. [17]. Distinguished by its pixel size of 30 m, each pixel corresponds to a ground area measuring 30 m by 30 m. This satellite operates with a temporal resolution of 16 days, enabling recurrent observations, and offers a spatial resolution of 15 m for panchromatic imagery and 30 m for multispectral imagery.

Fig. 1 shows the first four bands of Landsat-8 (Coastal/Aerosol, Blue, Green, and Red) which assume particular importance within the quasi-analytical algorithm employed for the estimation of water quality parameters [18]. Landsat-8, orbiting at an altitude of approximately 705 km, conducts its surveillance in a polar, sun-synchronous orbit, completing 14.5 orbits daily. This unique orbital inclination ensures that Landsat-8 passes over each terrestrial location consistently at the same local time of day. This orbital characteristic facilitates uniform illumination conditions and facilitates straightforward comparisons of image data over time. In addition to its multispectral capabilities, Landsat-8 is also equipped with hyperspectral data acquisition capabilities through its Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS) [19]. This facet enriches the satellite's utility by enabling the comprehensive study of Earth's surface composition and temperature.

3.2.1. Study area – River Ganges

Our key study area was centred on the critical and ecologically significant River Ganges as shown in Fig. 2. The Ganges, also known as the Ganga, is a major river in South Asia, primarily flowing through India and Bangladesh. The Ganges River originates in the western Himalayas in the Indian state of Uttarakhand and flows through several states in northern India, including Uttar Pradesh, Bihar, and West Bengal, before eventually entering Bangladesh and emptying into the Bay of Bengal. Water pollution in the River Ganges is a significant and longstanding environmental concern in India. The Ganges is not only one of the holiest rivers in the country, but it also serves as a lifeline for millions of people who rely on it for drinking water, irrigation, and various other purposes. However, the river has faced extensive pollution due to various human activities and industrial discharges (Rai, et al. [20], Yaseen [22]) We used a combination of remote sensing technologies and ground-based monitoring stations, to continuously gather water quality

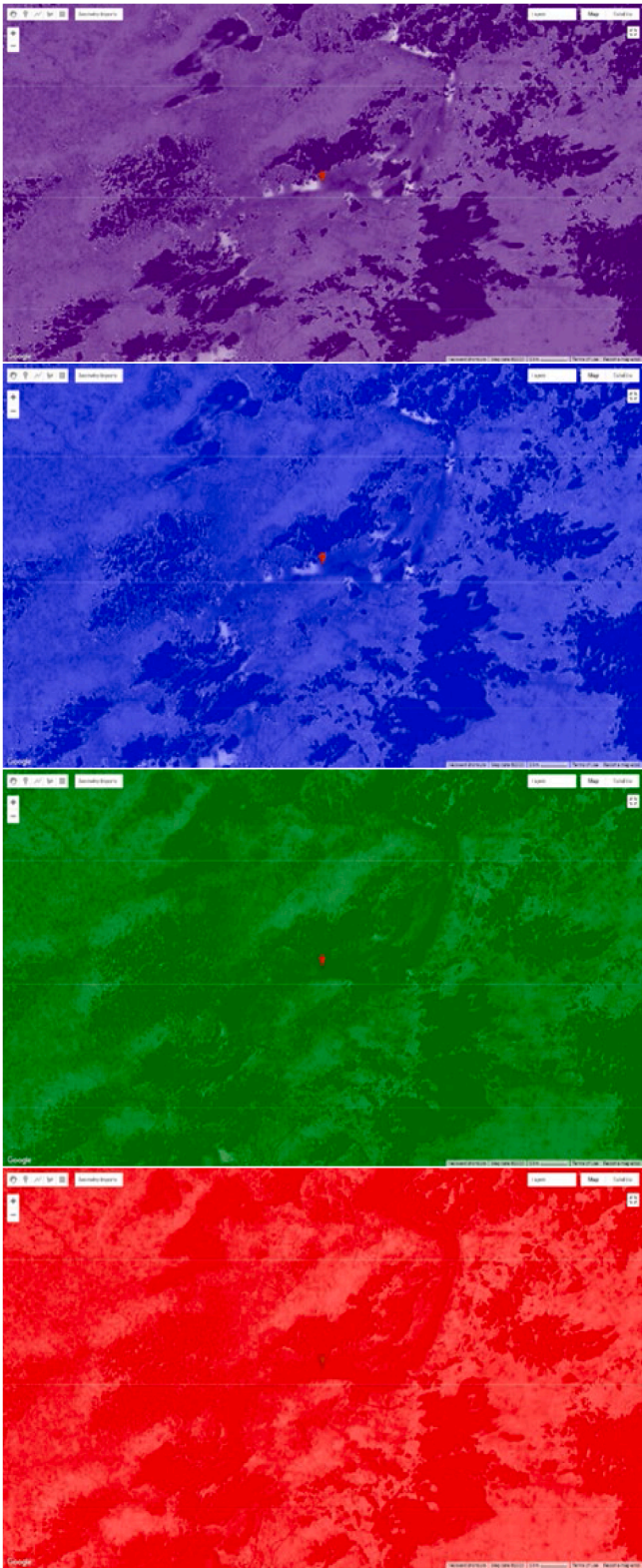


Fig. 1. Coastal, Blue, Green & Red bands of the landsat-8 image.

data. To achieve this, we utilized advanced data analysis techniques, comparing real-time water quality measurements to our extensive dataset. This approach enabled us to forecast and assess current and future water quality, identify trends, potential issues, and understand the impact of various factors on the river’s water quality.

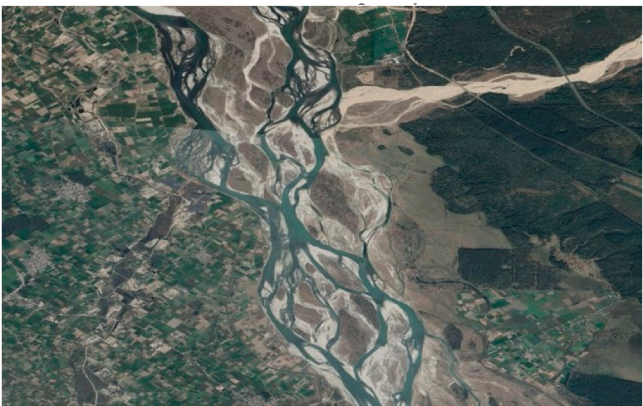


Fig. 2. Study area: river Ganges.

4. Methodology

In this section, we present the methodology employed in our novel approach to water quality monitoring and protection. Our system is designed to address the limitations of existing static data-based detection systems by incorporating dynamic data intake capabilities. This enables real-time identification of contamination sources and tracking of changes over time using remote sensing technology.

4.1. Phase I

The server system encompasses various components, notably the Google Earth Engine Data, the Quasi Analytical Algorithm, and real-time and time-series prediction models, and plays a pivotal role in processing the bulk of satellite data. The architecture is represented in Fig. 3. Through a mobile application interface, users can submit requests to the server, leveraging Google Earth Engine to access relevant satellite data. Subsequently, mathematical computations are applied to this data, and the outcomes are channelled into a pre-trained model for predictive analysis. The server also manages essential assets and intermediate files essential for prediction. This versatile system caters to a wide array of applications demanding real-time or time-series predictions, thereby offering substantial utility.

4.1.1. Satellite data

The proposed system’s API plays a specialized role in sourcing raw satellite data exclusively from Google Earth Engine, pinpointing the specified coordinates pertinent to Landsat 8. Among the extracted data, a salient parameter is the aquatic reflectance value, denoting the extent of light reflection by aquatic bodies encompassing lakes, rivers, and oceans. Landsat 8’s sensors exhibit exceptional precision in gauging

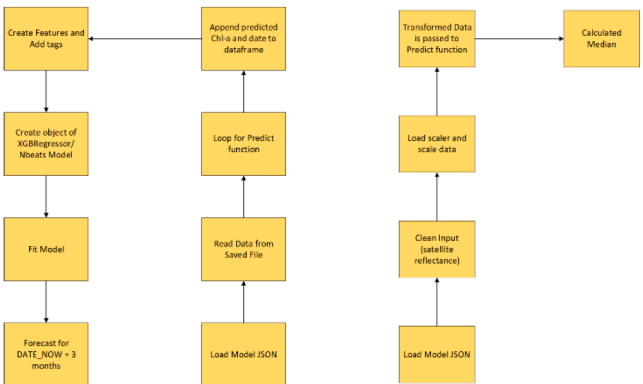


Fig. 3. Architecture.

water body reflectance, rendering this dataset invaluable for comprehensive investigations into diverse water body dynamics, spanning water quality assessment, sediment load assessment, and the identification of harmful algal blooms. Upon acquiring the Landsat 8 data, the API undertakes a series of intricate procedures, encompassing cloud cover mitigation and the extraction of reflectance values at specific wavelengths, such as Coastal, Red, Green, and Blue. These precise wavelength data are pivotal for subsequent computations employing the Quasi-Analytical Algorithm (QAA-v5). This algorithm proficiently calculates absorption and backscattering coefficients by utilizing the quartet of reflectance values. In doing so, it provides valuable insights into the optical attributes and related characteristics that define the aquatic environment.

4.1.2. Quasi-analytical algorithm

The Quasi-Analytical Algorithm (QAA v5) [21], stands as a prominent algorithm for extracting water quality parameters from satellite reflectance data, with a particular emphasis on determining chlorophyll-a concentrations in aquatic settings. This algorithm deploys a combination of empirical and semi-analytical methodologies to estimate chlorophyll-a concentrations from reflectance data.

The dynamic data acquisition system reliably identifies sources of pollution and distinguishes between natural and anthropogenic changes in water quality by integrating real-time satellite data with ground-based monitoring. By utilizing advanced algorithms, such as the Quasi-Analytical Algorithm (QAA-v5) and machine learning models, the system analyzes spectral reflectance and various water quality parameters. This analysis helps in recognizing patterns and anomalies associated with pollution events. The system's ability to continuously collect and process data allows it to differentiate between natural fluctuations, like seasonal changes, and pollution caused by industrial discharge or agricultural runoff. This comprehensive approach ensures accurate detection and attribution of water quality changes, enabling timely and targeted interventions.

The QAA v5 algorithm initiates by transforming reflectance data into remote-sensing reflectance (R_{rs}) through an intricate set of mathematical equations. Subsequently, employing a blend of empirical and semi-analytical techniques, the R_{rs} data is harnessed to derive estimates of chlorophyll-a concentrations. This estimation process incorporates a collection of coefficients and parameters, a product of in situ measurements. Furthermore, the algorithm adeptly rectifies the influence of other water constituents, including dissolved organic matter and coloured dissolved organic matter, relying on the estimated chlorophyll-a concentrations as reference points.

Comparative studies have revealed the commendable accuracy of the QAA v5 algorithm when juxtaposed with alternative methodologies. It has been demonstrated to attain chlorophyll-a concentration estimations with a precision rate ranging from 85 % to 90 %. Such a high level of accuracy renders the QAA v5 algorithm an invaluable instrument for the domains of water resource management and environmental surveillance. The QAA-v5 algorithm used in this paper is demonstrated below with all the required formulae.

- For, the ratio of the backscattering coefficient (b_b) to the sum of the backscattering and absorption coefficients ($b_b/(a+b_b)$) is calculated algebraically.

$$\frac{b_b(\lambda)}{a(\lambda) + b_b(\lambda)} = \frac{-0.0895 + \sqrt{0.008 + 0.499r_{rs}(\lambda)}}{0.249} \quad (1)$$

Here, r_{rs} is the spectral remote-sensing reflectance just beneath the surface calculated from nadir-viewing R_{rs} through:

$$r_{rs}(\lambda) = \frac{R_{rs}(\lambda)}{0.52 + 1.7R_{rs}(\lambda)} \quad (2)$$

- The widely used expression is used to model the spectral b_b .

$$b_b(\lambda) = b_{bw}(\lambda) + b_{bp}(\lambda_0) \left[\frac{\lambda_0}{\lambda} \right]^\eta \quad (3)$$

where b_{bw} is the backscattering coefficient of pure seawater and b_{bp} is the backscattering coefficient of suspended particles, respectively.

- When $a(\lambda_0)$, the ratio of $b_b/(a+b_b)$ at λ_0 , and $b_{bw}(\lambda_0)$ are known, $b_{bp}(\lambda_0)$ in Eq. (3) can be easily calculated using Eqs. (1) and (3). After estimating the power parameter (η), the values of b_b at other wavelengths are calculated.
- The total absorption coefficient at λ , $a(\lambda)$, is then calculated algebraically by applying $b_b(\lambda)$ to the ratio of $b_b/(a+b_b)$ at (Eq. (1)).
- After determining $a(\lambda)$, $a_{dg}(\lambda)$ and $a_{ph}(\lambda)$ are calculated using:

$$a_{dg}(443) = \frac{(a(411) - \zeta a(443)) - (a_w(411) - \zeta a_w(443))}{\xi - \zeta} \quad (4)$$

$$a_{ph}(\lambda) = a(\lambda) - a_w(\lambda) - a_{dg}(443)e^{-S(\lambda-443)} \quad (5)$$

- The constants required in the above equations are calculated as follows:

$$a(\lambda_0) = a_w(\lambda_0) + 10^{-1.146 - 1.366\chi - 0.469\chi^2} \quad (6)$$

$$\chi = \log \left(\frac{r_{rs}(443) + r_{rs}(490)}{r_{rs}(\lambda_0) + 3 \left(\frac{r_{rs}(667)}{r_{rs}(490)} \right) r_{rs}(667)} \right) \quad (7)$$

$$\eta = 2 \left(1 - 1.2e^{\left(-0.9 \frac{r_{rs}(443)}{r_{rs}(550)} \right)} \right) \quad (8)$$

$$\zeta = 0.74 + \frac{0.2}{0.8 + \frac{r_{rs}(443)}{r_{rs}(555)}} \quad (9)$$

$$\xi = e^{S(443-411)} \quad (10)$$

$$S = 0.015 + \frac{0.002}{0.6 + \frac{r_{rs}(443)}{r_{rs}(555)}} \quad (11)$$

- After Calculation all the values, we get the a_{ph} value at 440 wavelengths. That is used for calculating the chlorophyll value using the below formula:

$$CHL = \left[\frac{a_{ph}(440)}{0.05} \right]^{0.626} \quad (12)$$

Given the surface reflectance R_{rs} and the equation parameters, this calculates the chlorophyll value CHL and the ratio of the backscattering coefficient to the sum of the absorption and backscattering coefficients at a specified wavelength λ . Each parameter is iteratively calculated by the algorithm using equations that rely on earlier results. The algorithm outputs CHL and $b_b(\lambda)$ as its results. Because of the low

accuracy of the resulting chlorophyll value, not all these parameters are used.

4.2. Phase II

Following the computation of relevant parameters, a pre-trained regression model is employed to forecast the chlorophyll-a levels. Various machine learning and deep learning models underwent evaluation and testing to predict chlorophyll-a values, culminating in the selection of models that exhibited the highest degree of accuracy for final implementation.

Regression models such as XGBoost are widely adopted for such data analysis tasks owing to their ability to predict continuous values. These models operate by establishing an optimal-fit line or equation that correlates the input variables with the output variable, in this context, chlorophyll-a values. Subsequently, this equation is utilized by the model to generate predictions for novel, hitherto unobserved data.

CatBoost, a gradient boosting library that is exceptionally suited for handling categorical features and large datasets, emerges as a machine learning model well-suited to managing extensive datasets and resolving challenges characterized by numerous features. It operates as an implementation of gradient-boosting decision trees, effectively constructing an ensemble of elementary decision trees to facilitate predictions. CatBoost adopts an iterative approach, as pictorially shown in Fig. 4, by successively incorporating decision trees into the model and adjusting the trees' weights contingent on their performance.

In the context of chlorophyll-a value prediction, CatBoost is well-equipped to manage the extensive dataset replete with numerous features characterizing this task. The algorithm proficiently accommodates the non-linear associations between input variables and the output variable while simultaneously discerning the optimal combination of features for chlorophyll-a value prognostication.

CatBoost leverages gradient descent to minimize the loss function, progressively refining predictions over time. Furthermore, it incorporates a spectrum of advanced attributes, including built-in handling of categorical features, robustness to overfitting, and support for parallel processing, bolstering its capability to grapple with large datasets and high-dimensional predicaments. This makes CatBoost particularly suitable for tasks like chlorophyll-a value prediction where complex relationships and large datasets are common.

To address the potential issue of overfitting in the CatBoost model given the dynamic nature of water bodies and variability in chlorophyll-a concentration, we employed rigorous cross-validation techniques and monitored performance metrics across different data subsets. By splitting the dataset into training and validation sets, we ensured that the model's performance was evaluated on unseen data, helping to detect overfitting. Additionally, hyperparameter tuning was conducted to optimize the model complexity, and regularization techniques inherent in CatBoost were utilized to further mitigate overfitting. This comprehensive approach ensured that the CatBoost model generalized well to real-world data and maintained high accuracy in predicting chlorophyll-a concentrations across various conditions.

Given the reliance on a pre-trained CatBoost model for real-time

computations, the system incorporates new data and reflects changes in water quality parameters over time by continuously updating the pre-trained CatBoost model with fresh satellite and ground-based observations. As new data is collected, it is processed and used to fine-tune the model, ensuring that it remains responsive to current conditions. This dynamic updating mechanism allows the model to adapt to emerging trends and anomalies in water quality, maintaining high accuracy in its predictions. Additionally, the integration of time-series forecasting models like N-BEATS further enhances the system's ability to predict future water quality changes, providing a robust framework for ongoing water quality assessment and management.

CatBoost's ability to handle categorical features, optimize decision tree building, and efficiently handle large datasets makes it well-suited for predicting chlorophyll-a levels. Its ensemble-based approach, combined with gradient boosting and advanced optimization techniques, enables it to provide accurate and robust predictions for this specific challenge.

4.3. Phase III

The utilization of the N-BEATS (Non-linear and Basis function Ensemble for Time-series forecasting) model in the domain of water quality detection and time-series forecasting signifies a pivotal advancement in harnessing deep learning for environmental analysis. The application of N-BEATS has been instrumental in overcoming the challenges such as capturing the intricate short-term and long-term patterns inherent in chlorophyll-a concentration time series data. Additionally, N-BEATS has proven effective in handling the irregularities and fluctuations caused by external factors, such as environmental variations and human activities, that often pose challenges in accurate time-series forecasting.

One of the central attributes of the N-BEATS architecture as presented in Fig. 6 is its modularity, comprised of two foundational modules: the seasonal module and the trend module. Each module plays a distinctive role in capturing various temporal patterns. The seasonal module specializes in capturing transient patterns within the time series data, a critical factor when considering the impact of seasonality on water quality parameters. Meanwhile, the trend module is specifically designed to encapsulate enduring patterns, making it an invaluable tool for recognizing long-term variations that might be attributed to factors like climate change. This duality in the model's design enables it to effectively capture the full spectrum of temporal dynamics within the water quality dataset.

The crux of the N-BEATS model lies in its capacity to acquire a set of fundamental functions that serve as building blocks for predicting future values within the time series. These functions are leveraged to approximate virtually any time series, including the complex and intricate water quality data. By adopting these foundational functions, the model can make robust predictions for chlorophyll-a concentrations, accounting for both short-term fluctuations and long-term trends.

Parameter learning within the N-BEATS model is achieved through backpropagation, an iterative process in which the model continually refines its fundamental functions to minimize prediction errors. The training process is executed through a blend of mean squared error (MSE) and mean absolute error (MAE) metrics. Incorporating both Mean Squared Error (MSE) and Mean Absolute Error (MAE) metrics in the training process ensures a comprehensive evaluation of our models' performance, contributing to the accuracy and reliability of our predictions.

When predicting chlorophyll-a concentration, the N-BEATS model performs exceptionally well. It is particularly good at understanding complex patterns in data that changes over time. The seasonal module's capability to capture patterns influenced by seasonality, such as variations associated with different seasons or weather conditions, is indispensable for providing context to water quality predictions. Simultaneously, the trend module excels at encapsulating patterns

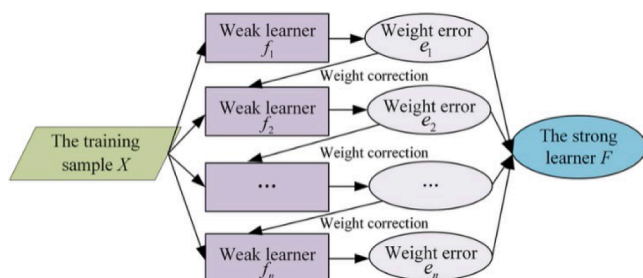


Fig. 4. CatBoost architecture.

influenced by more protracted variations, like those arising from climate change or gradual environmental shifts.

Furthermore, the N-BEATS model offers an enviable degree of configurability. This inherent flexibility enables researchers and environmental stakeholders to fine-tune the architecture to align precisely with the specific requirements of water quality analysis. This feature is instrumental in ensuring that the model's predictions are adaptable to diverse scenarios, accounting for the intricate interplay of variables affecting water quality.

In the context of this research, the N-BEATS model has demonstrated its proficiency in providing accurate and reliable predictions for chlorophyll-a concentrations. The high configurability, modular design, and capacity to learn from data make it a formidable tool in the domain of environmental analysis. By leveraging the deep learning capabilities of the N-BEATS model, the research can furnish precise forecasts for future chlorophyll-a values, enhancing our capacity to assess and manage water quality.

The integration of N-BEATS within the framework of water quality analysis serves as a beacon for future research, offering a promising avenue for enhancing the accuracy and comprehensiveness of environmental predictions. The application of deep learning and advanced time series forecasting models such as N-BEATS offers an exciting opportunity to usher in a new era of water quality assessment, providing invaluable insights for both researchers and environmental stakeholders.

Fig. 6 illustrates the modular architecture of N-BEATS, designed to detect patterns in time-series data. In the context of water quality, the N-BEATS model employs two core modules: a trend module and a seasonal module. The trend module identifies long-term changes in water quality, potentially indicating persistent environmental changes, while the seasonal module captures recurring patterns, such as seasonal variations in pollutant concentrations. This dual-module approach allows N-BEATS to correlate domain-specific variations in chlorophyll-a concentrations with temporal factors such as weather or seasonal fluctuations. By using domain-specific parameters (e.g., historical chlorophyll-a levels), the model is customized for accurate and meaningful water quality predictions.

5. Results

In this section, we present the results of our novel approach to water quality monitoring and protection, highlighting the performance metrics such as accuracy, precision, recall, Mean Absolute Error (MAE), and Mean Squared Error (MSE) of our system, which leverages the QAA-v5 algorithm, pre-trained CatBoost model, NBeats, and CatBoost Time-Series models as shown in the process flow diagram of our approach in Fig. 5, in accurately detecting and predicting chlorophyll-a concentration.

5.1. QAA-v5 algorithm accuracy

The outcomes of the QAA-v5 algorithm have been meticulously documented in Table 4, elucidating the percentage error and seven other regression metrics at each phase of the process. Given the intricacy of each operation entailed within the algorithm, a certain degree of inaccuracy is an inherent outcome. These inaccuracies were quantified utilizing the standard formula for percentage error and sklearn.metrics, affording a transparent evaluation of the error magnitude at every step.

The findings prominently underscore that the error in predicting chlorophyll-a values approaches nearly 100 percent, serving as a stark illustration of the limitations associated with relying solely on empirical methods. This emphasizes the imperative of incorporating alternative methodologies to enhance result accuracy. Pertaining to the integration of additional techniques, the table signifies that all parameters enlisted therein hold promise for assimilation into a deep learning model. An exception, however, emerges in the form of aph, which registers an error

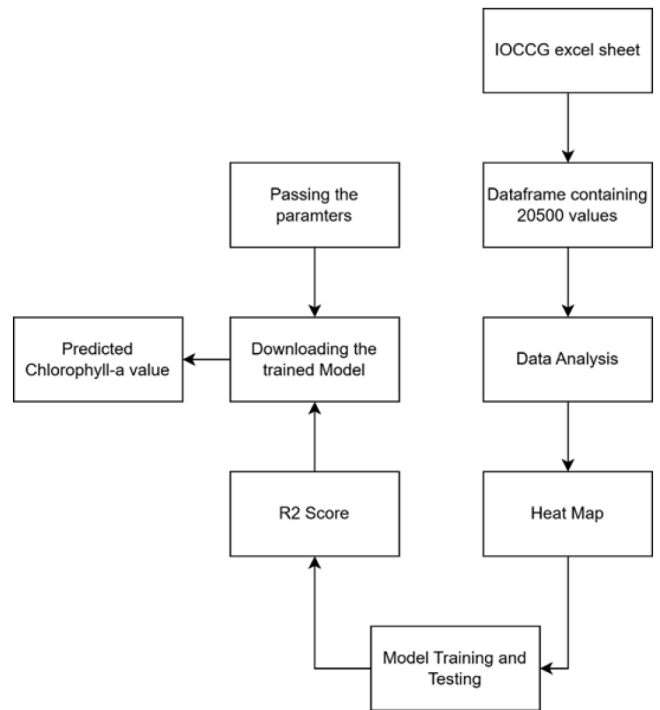


Fig. 5. Process Flow.

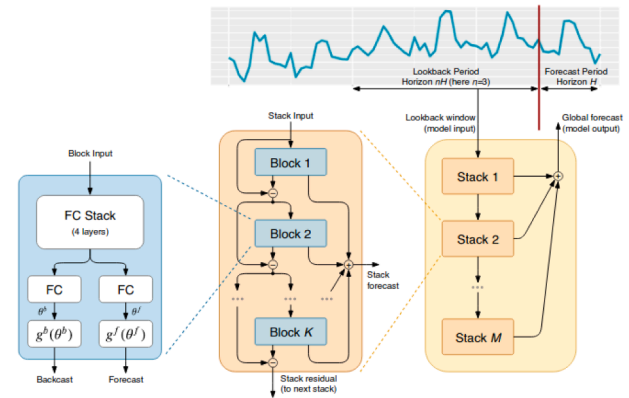


Fig. 6. Architecture of N-BEATS.

percentage of 178.27 as seen in Table 4. This observation imparts the necessity of judicious parameter selection when contemplating their incorporation into the model.

Consequently, for strengthening the model's resilience and precision, a semi-hybrid approach is posited as an advisable course of action. This hybridization entails the computation of specific values complemented by the application of deep learning models to provide supplementary support. This synergistic amalgamation of methodologies augments the prospects for heightened accuracy and reduced error in results, thereby endowing the model with enhanced reliability and efficacy.

5.2. IOCCG data analysis

Fig. 7 illustrates a correlation heatmap derived from the IOCCG dataset, offering a visual representation of the relationships between various parameters and chlorophyll-a. The comprehension of these correlations assumes paramount significance in the construction of an accurate model for chlorophyll-a prediction. The heatmap highlights the

λ	410	440	550	% Error	Regression Score	Mean Absolute Error	Mean Squared Error	Mean Absolute Percentage Error	Mean Squared Log Error	Mean Poisson Deviance	Mean Gamma Deviance
A	1.377	1.34	0.3762	23.14	0.865	0.1776	0.0373	0.2009	0.0076	0.0319	0.0348
Actual	1.5855	1.0882	0.3034								
Aph	0.163	0.421	1.077	178.27	-58.82	0.397	0.328	3.624	0.141	0.524	1.091
Actual	0.227	0.2724	0.0975								
Adg	1.536	0.913	0.136	3.957	0.995	0.0354	0.0013	0.0778	0.0004	0.0026	0.0135
Actual	1.588	0.939	0.1645								
Bb	0.0452	0.44	0.0417	13.333	-35.18	0.0051	0.00002	0.1331	2.491	0.0006	0.0152
Actual	0.0397	0.0381	0.0377								
Bbp	0.0418	0.0416	0.0408	14.26	-125.75	0.005	0.00002	0.142	0.00002	0.0006	0.0173
Actual	0.0364	0.0356	0.0367								
Rrs	0.0021	0.00295	0.0116	0	0.999	0.000006	1.3e-10	0.0022	1.3e-10	4.51e-8	0.000015
Actual	0.0021	0.00297	0.0116								
C	30.04	30.04	30.04	100.27	0.0	15.04	226.201	1.0026	0.439	9.245	0.387
Actual	15	15	15								

The observed lower correlation values between chlorophyll-a and 'a' and 'Rrs' should not be misconstrued as an indication of their insignificance in chlorophyll-a concentration determination. These parameters furnish indispensable insights that remain unattainable through other variables. Their omission from any predictive model would result in a substantial compromise in accuracy. Consequently, these parameters warrant classification as indispensable components in the development of models aimed at chlorophyll-a prediction.

The outcomes derived from the Recurrent Neural Network (RNN) model underscore its proficiency in predicting chlorophyll-a concentration. Demonstrating a robust regression score of 0.967, the model attains a commendable level of precision. Table 5 represents the CatBoost True and Predicted values for the samples. The selection of pertinent parameters, including 'bb,' 'wavelength,' 'Rrs,' 'r_rs,' 'a,' 'bb_dm,' and 'a_g,' for model training stems from their direct relevance to chlorophyll-a concentration.

The CatBoost model has proven to be an exceptionally effective solution for predicting chlorophyll-a concentration, exemplified by its impressive regression score of 0.9854. Table 6 represents the CatBoost True and Predicted values for the samples. The model underwent training utilizing the parameters 'bb,' 'wavelength,' 'Rrs,' 'r_rs,' 'a,' 'bb_dm,' and 'a_g,' meticulously chosen for their direct relevance to chlorophyll-a concentration. The model implementation featured the booster='gbtree' configuration and the objective='reg:squarederror' setting, accompanied by a learning rate of 0.01. A comprehensive training dataset comprising 20,500 samples played a pivotal role in refining the model's parameters and optimizing its performance. Fig. 9 graphically presents the CatBoost score and Analysis.

	1	0.83	-0.89	8.4e-17	8.6e-15	1.4e-14	-0.43	-0.4	-0.42	0.63	0.0046	-0.028	0.047	-0.49	-0.5
aw	0.83	1	-0.56	4.7e-16	7.5e-15	1.4e-14	-0.34	-0.23	-0.26	0.89	0.0042	-0.021	-0.031	-0.47	-0.48
bw	-0.89	-0.56	1	4.2e-16	6.5e-15	1.1e-14	0.43	0.48	0.48	-0.33	-0.0048	0.027	0.05	0.34	0.35
C	8.4e-17	4.7e-16	4.2e-16	1	-0.0067	-0.062	0.65	0.41	0.49	0.23	0.85	0.81	0.9	0.27	0.27
Sp	8.6e-15	7.5e-15	6.5e-15	-0.0067	1	-0.018	0.0029	-0.034	-0.012	-0.013	-0.028	0.04	0.017	0.042	0.041
Sd	1.4e-14	1.4e-14	1.1e-14	0.062	-0.018	1	-0.044	-0.01	-0.073	-0.012	-0.069	-0.076	-0.08	-0.033	-0.033
a.ph	-0.43	-0.34	0.43	0.65	0.0029	-0.044	1	0.71	0.78	0.049	0.53	0.55	0.6	0.18	0.19
a.g	-0.4	-0.23	0.48	0.41	-0.034	-0.01	0.71	1	0.75	0.24	0.3	0.33	0.36	-0.04	-0.039
a.dm	-0.42	-0.26	0.48	0.49	-0.012	-0.073	0.78	0.75	1	0.14	0.4	0.39	0.44	0.072	0.073
a	0.63	0.89	-0.33	0.23	-0.013	-0.012	0.049	0.24	0.14	1	0.18	0.17	0.18	-0.47	-0.48
bb.ch	0.0046	0.0042	-0.0048	0.85	-0.028	-0.069	0.53	0.3	0.4	0.18	1	0.65	0.85	0.31	0.31
bb.dm	-0.028	-0.021	0.027	0.81	0.04	-0.076	0.55	0.33	0.39	0.17	0.65	1	0.96	0.39	0.39
bb	-0.047	-0.031	0.05	0.9	0.017	-0.08	0.6	0.36	0.44	0.18	0.85	0.96	1	0.4	0.4
Prs	-0.49	-0.47	0.34	0.27	0.042	-0.033	0.18	-0.04	0.072	-0.47	0.31	0.39	0.4	1	1
r.rs	-0.5	-0.48	0.35	0.27	0.041	-0.033	0.19	-0.039	0.073	-0.48	0.31	0.39	0.4	1	1

Sr. No.	True Value	Predicted Value
1	10	10.0059
2	0.5	0.940024
3	0.2	0.340044
4	2	3.032543
5	0.1	0.275102
6	15	14.55523
7	5	2.638203
8	0.5	0.640473
9	0.03	0.227554
10	25	28.13308
11	0.05	0.250235
12	7	2.756667
13	0.7	1.729146
14	20	7.227631
15	0.05	0.289587
16	2	1.633988
17	1.5	2.001487
18	20	15.09259
19	10	6.491951
20	2	2.468056



Fig. 8. RNN R2 score and Analysis (50 random values).

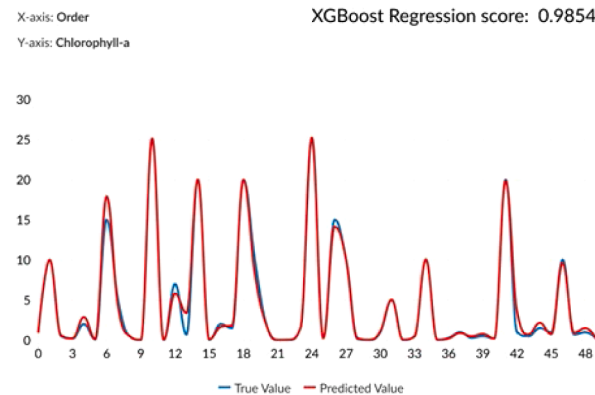


Fig. 9. CatBoost R2 score and Analysis (50 random values).

Table 6
CatBoost true & predicted value (20 random values).

Sr. No.	True Value	Predicted Value
1	10	9.989175797
2	0.5	0.614284337
3	0.2	0.235272571
4	2	2.870461941
5	0.1	0.159304753
6	15	17.90938568
7	5	4.178232193
8	0.5	0.534790337
9	0.03	0.042925667
10	25	25.11787987
11	0.05	0.04803139
12	7	5.79621315
13	0.7	3.387304068
14	20	20.01626968
15	0.05	0.103164397
16	2	1.657328606
17	1.5	1.85707593
18	20	19.97551346
19	10	8.215953827
20	2	1.941943169

Secondly, CatBoost boasts high scalability and easy parallelization capabilities, rendering it particularly apt for large-scale datasets.

Furthermore, CatBoost offers a multitude of hyperparameters that can be meticulously tuned to enhance its performance. Parameters such as the learning rate, tree quantity, and tree depth can be optimized to align with the specific requirements of the data, ensuring optimal model performance. Moreover, CatBoost is renowned for its high interpretability, simplifying the process of comprehending and elucidating its outcomes.

5.5. Model evaluation

The RNN model demonstrates strong predictive capabilities with an R2 score of 0.967, signifying that it can explain approximately 96.7 % of the variance in the data as represented in Table 7. It surpasses the CatBoost model when considering the Mean Absolute Error (0.81) and Mean Squared Error (2.64). Additionally, the RNN model exhibits a mean absolute percentage error of 0.49 and a mean squared log error of 0.047. Further evaluation reveals a mean Poisson Deviance of 0.35 and a mean Gamma Deviance of 0.29. In contrast, the CatBoost model excels in terms of the R2 score, boasting a value of 0.985. Moreover, it outperforms the RNN model in both Mean Absolute Error and Mean Squared Error, with values of 0.45 and 1.17, respectively. The CatBoost model exhibits a mean absolute percentage error of 0.29 and a mean squared log error of 0.023. It also demonstrates a mean Poisson Deviance of 0.16 and a mean Gamma Deviance of 0.13. Overall, when

assessing model performance, the CatBoost model outshines the RNN model, displaying a higher R2 score and superior performance across various error metrics.

5.6. Real-time analysis of CatBoost model

Model accuracy serves as a vital metric for assessing its performance and suitability for a specific task. It is imperative to acknowledge that accuracy alone can be deceptive, potentially overestimating a model’s capabilities. This overestimation can be attributed to various factors, such as training the model with non-representative data, utilizing data from a single location or condition during training, or succumbing to overfitting, leading to an inflated perception of its performance. To overcome these limitations and provide a more reliable assessment of the model’s performance, we conducted a real-time analysis of water quality using Landsat satellite data. This approach allowed for a comprehensive and real-time evaluation of water quality, enabling continuous monitoring and timely detection of temporal changes. Moreover, the model’s accuracy was validated using NOAA’s chlorophyll dataset, affording a robust appraisal of its capabilities.

The presented results, as tabulated in Table 8, affirm that the model’s accuracy met the required standards, demonstrating its applicability to a diverse range of inland water bodies. This degree of accuracy is commendable for a remote sensing solution designed for water quality analysis. Additionally, it’s noteworthy that the predicted values and remote sensing images utilized in this analysis were obtained on January 17th, 2023, underscoring the model’s provision of precise, up-to-date information—an indispensable asset for stakeholders reliant on dependable data.

The line graph in Fig. 10 further visually accentuates the comparison between actual and predicted chlorophyll-a values in real-time. The graph distinctly showcases the model’s performance, with the true and predicted value curves closely aligning. This concordance substantiates the model’s adeptness at accurately forecasting chlorophyll-a values in real-time. Thus, the graph offers a lucid representation of the model’s efficacy.

As shown in Table 9 with an R2 score of 0.72, this model accounts for 72.2 % of the variance within the dataset. The model exhibits a Mean

Table 7
Model evaluation metrics.

Model	RNN	CatBoost
Regression Score	0.967	0.985
Mean Absolute Error	0.812	0.453
Mean Squared Error	2.635	1.169
Mean Absolute Percentage Error	0.488	0.291
Mean Squared Log Error	0.0467	0.0228
Mean Poisson Deviance	0.354	0.1602
Mean Gamma Deviance	0.286	0.1268

Table 8
Real-time analysis of CatBoost model.

Place	Lat	Long	True Value	Predicted Value	% Error
Qinghai Lake	36.95625	100.06875	2.814443	2.8907325	2.710642923
Mumbai	20.00625	72.01875	3.3938935	3.42616	0.950722231
Pulicat Lake	14.00625	80.45625	0.27339086	0.8084742	195.7210054
Krishna River	16.51875	79.1815	3.3497884	5.2188725	55.7970796
Tapi River	21.99375	72.28125	4.8934975	5.8690114	19.93490137
Sabarmati River	22.14375	72.3563	6.87	5.907296	14.01315866

Absolute Error of 0.74 and a Mean Squared Error of 0.94. The mean absolute percentage error for this model is 0.23, and the mean squared log error is 0.049. The model's Mean Poisson Deviance and Mean Gamma Deviance are 0.32 and 0.34, respectively. Overall, the model's performance is reasonable, but there is certainly room for improvement. The R2 score of 0.72 suggests that the model accounts for 72.2 % of the variance in the dataset. While this indicates a reasonable level of performance, the relatively high values for the error metrics raise concerns about the accuracy of the model's predictions. Specifically, the Mean Absolute Percentage Error of 0.23 highlights an average deviation of approximately 23 % in the model's predictions from the actual values. Furthermore, the Mean Squared Log Error suggests a potential underestimation of some higher values within the dataset. These observations indicate that there is room for improvement in the model's accuracy and predictive capabilities.

5.7. Future timeseries analysis

The utilization of the N-Beats algorithm for time series analysis in this research is noteworthy. NBeats is an advanced model designed for predicting future values in time series data. It achieves this through a stacked architecture comprising fully connected layers, attention mechanisms, and gated recurrent units. This algorithm's suitability for time series analysis of water quality is evident due to its capacity to handle intricate and long-term dependencies, which are frequently encountered in environmental data. In the context of water quality analysis, multivariate time series analysis is essential as it considers various environmental variables like temperature, precipitation, and pollution levels, all of which influence water quality. NBeats' ability to simultaneously consider multiple variables positions it as a robust choice for assessing water quality data.

Nevertheless, predicting future water quality based solely on time remains a formidable challenge. External factors, such as human activities, can exert a substantial impact on water quality. The inherent complexity of this task inevitably affects the accuracy of predictions. While the accuracy of NBeats in this study is reasonably satisfactory, accurately forecasting future water quality levels as represented in Table 10 remains a demanding endeavour.

The line graph in Fig. 11, depicting the true and predicted values of

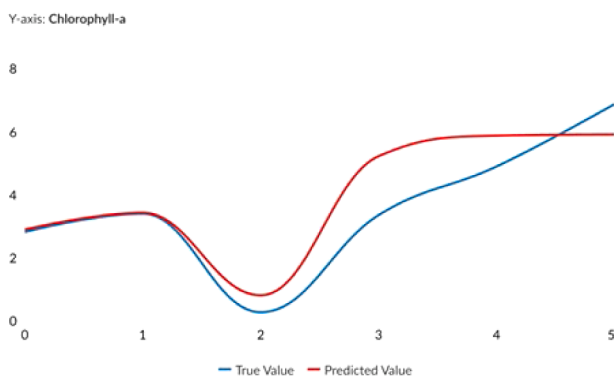


Fig. 10. Real-Time Analysis of CatBoost Model.

Table 9
CatBoost evaluation metrics (Real time analysis).

Regression Score	0.721
Mean Absolute Error	0.741
Mean Squared Error	0.944
Mean Absolute Percentage Error	0.2308
Mean Squared Log Error	0.0486
Mean Poisson Deviance	0.316
Mean Gamma Deviance	0.338

chlorophyll-a lends support to the effectiveness of N-Beats in predicting future water quality levels. It's worth noting that the data used pertains to the Ganges River in the city of Patna.

6. Discussion

In the realm of water quality assessment and analysis, the need for models capable of providing real-time, reliable results cannot be overstated. Historically, empirical formulas have been a common approach; however, this research revealed their limitations in terms of accuracy. To address this challenge, we embarked on an exploration of hybrid models, harnessing the strengths of multiple models to craft a superior solution. The outcomes of our experimentation were profoundly encouraging, with the hybrid model achieving an exceptional accuracy rate of 98.5 % for the static dataset. This represents a substantial leap in accuracy compared to the empirical formula.

The reported overall accuracy of 98.5 % refers to the model's performance across the entire dataset, demonstrating its general predictive strength. However, individual cases such as Pulicat and Krishna Lakes exhibit higher error rates (195 % and 55 %, respectively), highlighting potential model limitations in areas with unique environmental characteristics or data discrepancies. These instances underscore the complexity of achieving uniformly high accuracy across diverse water bodies. Moving forward, we aim to address such discrepancies by further tuning model parameters and incorporating localized data adjustments, which would help reduce prediction errors in outlier locations.

A pivotal advantage of the hybrid model lies in its adaptability to real-time data. Our experiments demonstrated that the model exhibited versatility, devoid of regional or conditional constraints, making it universally applicable to inland water bodies worldwide. This adaptability renders it an ideal choice for stakeholders seeking dependable, real-time insights into water quality.

Forecasting future water quality levels is an intricate undertaking, as numerous variables can influence water quality. Nevertheless, our research results underscore the hybrid model's capability to furnish

Table 10
Future timeseries analysis.

Sr. No.	True Value	Predicted Value
1	9.279606819	10.13039303
2	6.313136578	7.4802618
3	7.436928272	9.4510479
4	9.954713821	10.7449789
5	9.275117874	11.51153183
6	13.95747948	13.86681747

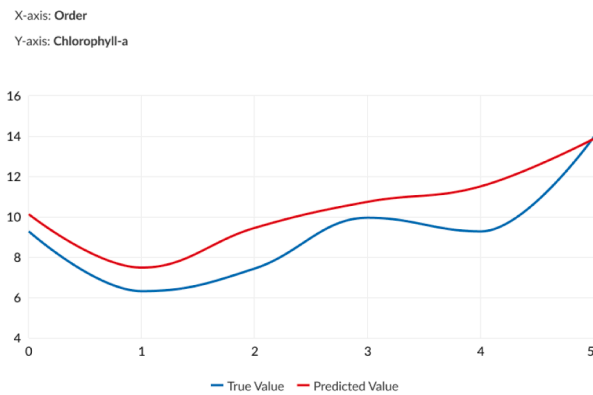


Fig. 11. Future timeseries analysis of NBEATS model.

precise and pertinent predictions. The integration of NBeats, an advanced algorithm tailored for time series analysis, played a pivotal role in ensuring the accuracy of these predictions.

Despite some inherent limitations, such as the challenge of acquiring accurate imagery under cloudy conditions, the advantages of employing remote sensing technology for water quality analysis are compelling. Notably, remote sensing technology offers a more cost-effective alternative to traditional methods of water quality analysis. Moreover, our research demonstrated that the hybrid model performed admirably even under clear sky conditions, affirming its effectiveness.

7. Conclusion

Our work underscores the promising future of employing remote sensing and hybrid models for water quality analysis. Our study introduces a new hybrid model that was synergized with established algorithms by merging deep learning innovations with a flexible approach to water quality analysis for global applications. The model developed in this paper, based on the rich IOCCG dataset, significantly extends existing methods by showing that remote sensing could achieve a much higher level of precision for both static and real-time water quality assessment. No attempt has been made so far to come up with an approach which is adaptable and would thus be able to bypass the limitation of region-specific algorithms, something that so far has constrained this line of research.

That is one of the great results of the model: it is robust, cheap, even in conditions that are not very favorable, such as cloud cover—a classic problem with satellite imaging. This model's capability extends far beyond its excellent accuracy into being eminently promising for real-time adaptability and a game-changer for proactive water quality management.

Therefore, this system can be taken to the next level of sophistication by including data from many satellites and building deep learning models that are specifically optimized for remote sensing applications. This will, of course, work for greater precision and wider applicability whereby the hybrid approach can sit in the vanguard of water quality analyses across diverging environmental conditions worldwide.

Approval of the submitted version of the manuscript

Please check this box to confirm that all co-authors have read and approved the version of the manuscript that is submitted. Signatures are not required.

CRediT authorship contribution statement

Lavanya Kandasamy: Data curation, Conceptualization. **Anand Mahendran:** Methodology, Formal analysis. **Sai Harsha Varma Sangaraju:** Validation, Software. **Preksha Mathur:** Visualization, Software.

Soham Vijaykumar Faldu: . **Manuel Mazzara:** Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] P. Futó, E. Lengyel, M. Futó, Z. Németh, Z. Pirger, A. Komáromy, G. Bernát, Ecophysiological characterisation of a *Klebsormidium* strain isolated from a cave environment, *J. Appl. Phycol.* 36 (3) (2024) 1197–1208.
- [2] B.M.D.M. Guimarães, I.E.L. Neto, Chlorophyll-a prediction in tropical reservoirs as a function of hydroclimatic variability and water quality, *Environ. Sci. Pollution Res.* 30 (39) (2023) 91028–91045.
- [3] J.K. Summers (Ed.), *Water Quality: Science, Assessments and Policy*, BoD—Books on Demand, 2020.
- [4] M. Elhag, I. Gitas, A. Othman, J. Bahrawi, P. Gikas, Assessment of water quality parameters using temporal remote sensing spectral reflectance in arid environments, Saudi Arabia. *Water* 11 (3) (2019) 556.
- [5] T.M. Olatunde, F.A. Adelani, Z.Q. Sikhakhane, A review of smart water management systems from Africa and the United States, *Eng. Sci. Technol. J.* 5 (4) (2024) 1231–1242.
- [6] H. Yang, J. Kong, H. Hu, Y. Du, M. Gao, F. Chen, A review of remote sensing for water quality retrieval: progress and challenges, *Remote Sens. (Basel)* 14 (8) (2022) 1770.
- [7] M.W. Matthews, Bio-optical modeling of phytoplankton chlorophyll-a. *Bio-optical Modeling and Remote Sensing of Inland Waters*, Elsevier, 2017, pp. 157–188.
- [8] T. Chen, C. Guestrin, Xgboost: a scalable tree boosting system, in: *Proceedings of the 22nd acm sigkdd international conference on knowledge discovery and data mining*, 2016, pp. 785–794.
- [9] Q. Ye, X. Yang, C. Chen, J. Wang, River water quality parameters prediction method based on LSTM-RNN model, in: *2019 Chinese Control And Decision Conference (CCDC)*, IEEE, 2019, pp. 3024–3028.
- [10] B.N. Oreshkin, D. Carпов, N. Chapados, Y. Bengio, N-BEATS: neural basis expansion analysis for interpretable time series forecasting, *arXiv preprint (2019) arXiv:1905.10437*.
- [11] N. Nasir, A. Kansal, O. Alshaltone, F. Barneih, M. Sameer, A. Shanableh, A. Al-Shamma'a, Water quality classification using machine learning algorithms, *J. Water Process Eng.* 48 (2022) 102920.
- [12] M.G. Uddin, S. Nash, A. Rahman, A.I. Olbert, Performance analysis of the water quality index model for predicting water state using machine learning techniques, *Process Saf. Environ. Prot.* 169 (2023) 808–828.
- [13] P. Dutta, R. Chaki, A survey of data mining applications in water quality management, in: *Proceedings of the CUBE International Information Technology Conference*, 2012, pp. 470–475.
- [14] A. Dhumvad, S. Prabhu, S.F. Da'Silva, S. Simu, P. Padiyar, V. Turkar, V. Salgaonkar, Water pollution monitoring and decision support system, in: *2022 3rd International Conference for Emerging Technology (INCET)*, IEEE, 2022, pp. 1–6.
- [15] X. Sun, Y. Zhang, K. Shi, Y. Zhang, N. Li, W. Wang, B. Qin, Monitoring water quality using proximal remote sensing technology, *Sci. Total Environ.* 803 (2022) 149805.
- [16] M.T. Duy, Monitoring of the Quality of Marine Waters in French Guiana By Remote Sensing and In-Situ Measurements, June, Université du Littoral Côte d'Opale, 2023.
- [17] J. Tavora, B. Jiang, T. Kiffney, G. Bourdin, P.C. Gray, L.S. de Carvalho, E. Boss, Recipes for the derivation of water quality parameters using the high-spatial-resolution data from sensors on board sentinel-2A, sentinel-2B, Landsat-5, Landsat-7, Landsat-8, and Landsat-9 satellites, *J. Remote Sens.* 3 (2023) 0049.
- [18] L. Maslukah, M. Sadewo, B. Rochaddi, S. Widada, A. Wirasatriya, U.J. Wisha, Estimation of suspended particulate matter using landsat 9 imagery: generating algorithms and spatio-temporal distributions, *ILMU KELAUTAN: Indonesian J. Marine Sci.* 29 (1) (2024).
- [19] H. Dibs, A.H. Ali, N. Al-Ansari, S.A. Abed, Fusion Landsat-8 thermal TIRS and OLI datasets for superior monitoring and change detection using remote sensing, *Emerg. Sci. J.* 7 (2) (2023) 428–444.
- [20] P. Rai, J.Y. Kumar, A. Sen, Ganga, GAP, and lockdown: potential threats to the biodiversity of the river, *J. Water and Climate Change* (2024) jwc2024322.
- [21] J.A. Schulien, T. Code, C. DeGasperi, D.A. Beauchamp, A. Tonus Ellis, A.H. Litt, Annual and interannual variability in the diffuse attenuation coefficient and turbidity in urbanized washington lake from 2013 to 2022 assessed using landsat-8/9, *Remote Sens. (Basel)* 15 (20) (2023) 5055.
- [22] Z.M. Yaseen, Flood hazards and susceptibility detection for Ganga river, Bihar state, India: employment of remote sensing and statistical approaches, *Results Eng.* 21 (2024) 101665.



Dr. K. Lavanya is a distinguished academician and researcher currently serving as a Professor in the School of Computer Science and Engineering (SCOPE) at VIT Vellore. Her current research interests encompass a range of topics within the fields of Computational Intelligence, Data Science, and Machine Learning. She is particularly interested in the application of NoSQL databases, data mining, and data warehousing techniques to extract meaningful insights from large and complex datasets. She is a prolific contributor to the scientific community, having authored over 30 research papers that have been published in prestigious international journals and presented at renowned conferences.



Anand Mahendran is working as a Professor in School of Computer Science and Engineering, Vellore Institute of Technology (VIT), Chennai, India. He received his Ph.D.(Computer Science and Engineering) from Vellore Institute of Technology (VIT), Vellore, India, in 2012, M.E.(Computer Science and Engineering) from Anna University, India, in 2005, and his B.E. (Computer Science and Engineering) from Madras University, India, in 2003. His research interests include formal language and automata theory, bio-inspired computing models. He has published >75 papers in international journals and refereed international conferences. Earlier, he worked as a Post-Doctoral Research Fellow in Laboratory of Theoretical Computer Science, National Research University, Higher School of

Economics(HSE), Moscow, Russia.



Sai Harsha Varma Sangaraju was born in 2002. He is currently pursuing a Master of Science in Computer Science at New York University, New York, USA, and earned his Bachelor of Technology in Computer Science and Engineering from Vellore Institute of Technology (VIT), Vellore, India, in 2024. His research interests lie in Artificial Intelligence, Machine Learning, Natural Language Processing, and Deep Learning. He has published multiple research papers in reputable conferences and journals, focusing on diverse domains such as environmental science, healthcare, and engineering. He is dedicated to leveraging computational methods and cutting-edge technologies to address complex real-world challenges, with a strong focus on advancing the frontiers of AI and data-

driven innovations.



Preksha Mathur, born in 2002, is a Master's student in Computer Science at the University of California, Riverside. She graduated with a B.Tech. in Computer Science and Engineering from Vellore Institute of Technology (VIT), Vellore, India, in 2024. Her academic journey reflects a deep passion for cutting-edge technologies, focusing on Machine Learning, Deep Learning, Natural Language Processing, and Image Processing. As an emerging researcher, Preksha is dedicated to advancing the frontiers of computer science.



Soham Faldu is a versatile Software Developer with balanced expertise in software development, app development, and machine learning. Proficient in Flutter and various programming languages, he specializes in building high-performance applications and integrating machine learning into user-focused solutions. He is passionate about leveraging emerging technologies, including quantum computing and cryptography, to drive innovation.



Manuel Mazzara is a Professor of Computer Science at Innopolis University (Russia) with a research background in Software Engineering, Service-Oriented Architecture, concurrency theory, formal methods, and software verification. He published many relevant and highly cited papers, in particular in the field of Service Engineering and Software Architectures. Manuel received a Ph.D. in computing science from the University of Bologna, Italy, and has collaborated with European and US industries, plus governmental and inter-governmental organizations such as the United Nations, always at the edge between science and software production. The work conducted by Dr. Mazzara and his team in recent years focuses on the development of theories, methods, tools, and programs covering the two major aspects of Software Engineering: the process side, related to how we develop software, and the product side, concerning the results of this process. Currently, Dr. Mazzara is the director of the Institute of Software Development and Engineering and the Head of the International Cooperation Office at Innopolis University.